Saran S

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Summary

Integrated M.Tech Software Engineering student at VIT Chennai with hands-on experience in software development, full-stack applications, and REST API integration. Strong foundation in Python, Java, Data Structures & Algorithms, and object-oriented programming, with experience building backend systems and data-driven applications.

Education

M.Tech Integrated – Software Engineering | 7.79/10 CGPA 2022 – 2027
Vellore Institute of Technology, Chennai

12th – HSC | 92.33% 2021 – 2022
Dr. Natarajan Matriculation Higher Secondary School

10th – SSC | 99.8% 2019 – 2020
Dr. Natarajan Matriculation Higher Secondary School

Technical Skills

Programming Languages: Python, Java, JavaScript, SQL
Core CS: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems
Web & Backend: REST APIs, FastAPI, Next.js, PostgreSQL, MySQL
Data & AI: Pandas, NumPy, Scikit-learn, XGBoost, RAG, Embeddings
Tools & Practices: Git, GitHub, Agile/Scrum, SDLC

Experience

ASP RCM Solutions Pvt. Ltd. June 2025 – July 2025
Software Development Intern

- Developed responsive frontend components for a data-driven application, improving usability and UI consistency.
- Integrated REST APIs to retrieve, manage, and display application data across frontend workflows.
- Optimized SQL queries to improve data retrieval performance and application efficiency.
- Designed reusable UI components to improve code maintainability and consistency across application modules.

Projects

AI-Powered Equity Analysis Platform

- Developed a full-stack platform for company analysis, portfolio tracking, and investment research.
- Implemented a retrieval-augmented generation (RAG) chatbot using embeddings and vector search for context-aware company research.
- Integrated authentication, watchlists, holdings tracking, and financial data into responsive dashboards.
- **Technologies: Next.js, FastAPI, PostgreSQL, Gemini AI, Qdrant**

Adaptive SQL Query Performance Predictor

- Processed historical SQL query execution logs using Python and Pandas to prepare data for analysis.
- Extracted query-level features and trained an XGBoost model to predict SQL query execution performance.
- Developed a prediction pipeline to identify potentially slow-running queries and support database performance analysis.
- **Technologies: Python, PostgreSQL, XGBoost, Pandas, NumPy**

Loan Default Risk Prediction System

- Developed a machine learning classification system to predict loan default risk using applicant and loan characteristics.
- Preprocessed financial data and engineered features for model training and prediction.
- Implemented a Logistic Regression pipeline to classify applicants according to predicted default risk.
- **Technologies: Python, Pandas, NumPy, Logistic Regression, FastAPI**

Leadership

Arignar Anna Thamizh Mandram – VIT 2024 – 2025
Joint Secretary

- Led teams and coordinated cultural events through effective planning, communication, and execution.
- Managed event coordination and collaborated with student teams to ensure smooth program execution.

Student Welfare – VIT 2023 – 2024
Student Outreach Team Member

- Collaborated with diverse teams to plan and organize student outreach initiatives, ensuring smooth coordination.
- Strengthened communication, teamwork, and community engagement through hands-on involvement in outreach activities.